Chapter 11. Mediation, SWIGs and G-Methods

Direct and indirect effects, single-world intervention graphs, the g-formula and its estimators

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11.1 Direct and indirect effects

Motivation. The central question is how to decompose a total causal effect into the part transmitted directly and the part transmitted through mediators, in nonlinear models where "holding the mediator fixed" and "subtracting" no longer give a well-defined indirect effect.

11.1.1 Preliminaries: potential responses

Motivation. Fix counterfactual notation, including nested counterfactuals.

In an SCM, $Y(u)$ is the value of Y for unit $U = u$ (the potential response). $Y_x(u)$ is the value of Y for the same unit under $\text{do}(X = x)$, so $P(Y_x = y) = P(y \mid \text{do}(x))$. $Y_{xz}(u)$ is the value under the joint intervention $\text{do}(X = x, Z = z)$. $Z_{x^*}(u)$ is the mediator value that would naturally occur under the reference intervention $\text{do}(X = x^*)$. The nested counterfactual

$$Y_{x,Z_{x^*}}(u): \quad \text{set } X = x, \text{ but set } Z \text{ to the value it would have taken under } X = x^*.$$

11.1.2 Introduction

Motivation. Motivate direct and indirect effects and the prescriptive/descriptive distinction.

For $X \to Z \to Y$ with $X \to Y$:

- **Total effect:** the change in the distribution of Y when X is set to x by external intervention, $P(Y_x = y) = P(y \mid \text{do}(x))$.
- **Direct effect:** measured by holding all intermediate variables constant.
- **Indirect effect:** in a linear system, indirect = total − direct; in a nonlinear system it cannot be measured by holding some variables constant.

Informally, a direct effect quantifies an influence not mediated by other variables; more formally, the sensitivity of Y to changes in X while all other factors in the analysis are held fixed. Graphically, holding those factors fixed eliminates all causal paths from X to Y except the direct link $X \to Y$. Indirect effects cannot be defined this way: no choice of variables to hold constant makes the measured effect circumvent the direct pathway. Hence the definition of indirect effects remained incomplete.

11.1.2.1 Examples

Birth-control pill and thrombosis. The pill may increase thrombosis directly but reduce it indirectly by lowering pregnancy rates (pregnancy itself increases thrombosis). The direct effect is biologically stable and invariant to social factors (e.g. marital status), more transportable and more useful in scientific explanation and policy analysis than the total effect.

Race/sex discrimination in hiring. Courts care about the direct effect of race or gender on hiring, not the indirect effect via qualifications. The legal test: would the employer have made the same decision had the applicant's race or gender been different, all else being equal?

Smoking, hypertension and heart disease (Hernán and Robins 2020, Ch. 23). A randomized trial finds that quitting smoking ($A = 0$) reduces myocardial infarction Y. Does the benefit operate through reduced hypertension M (measured at 6 months)? Direct effect: $A \to Y$ not through M (oxidative stress, platelet effects); indirect effect: $A \to M \to Y$. Policy question: if a drug eliminated smoking-induced hypertension, how much of the MI benefit would remain?

Drug, aspirin and headache. A drug has a side effect, headache; patients with headaches tend to take aspirin, which may affect the disease directly or modify the impact of the drug. The total effect reflects existing aspirin usage patterns; the direct effect isolates the drug's impact regardless of aspirin intake. A direct effect such as

$$P(Y_{xz} = y) - P(Y_{x^*z} = y),$$

with z a specified aspirin level, informs whether aspirin should be encouraged or discouraged during treatment. In linear systems direct effects are given by path coefficients and do not depend on the level at which intermediates are held; in nonlinear systems that level generally modifies the effect and must represent the target policy. This leads to two conceptualisations.

11.1.2.2 Prescriptive versus descriptive

- **Prescriptive (controlled direct effect):** would an untreated patient improve if treated, while aspirin intake is held at a predetermined level $Z = z$? It does not rely on individual natural behavior: $P(Y_{xz} = y) - P(Y_{x^*z} = y)$.
- **Descriptive (natural direct effect):** would the untreated patient improve if treated, while aspirin intake is held at the level the patient currently consumes under no treatment? It requires knowledge of individual natural behavior.

Consider a patient who takes aspirin iff treated, and for whom treatment works only with aspirin. Then the natural direct effect is zero, whereas the controlled direct effect (at $z =$ aspirin) is non-zero. For population averages: the prescriptive formulation is *pragmatic* (predict the difference in recovery when a prescribed aspirin dose is given to everyone), while the descriptive formulation is *attributional* (is an observed improvement due to the treatment itself, as opposed to preferential aspirin use among the treated?).

Why the average natural direct effect? Suppose the manufacturer considers eliminating the headache side effect. Would the drug retain its effectiveness in the population? The CDE does not answer this, since it refers to one aspirin level imposed uniformly. In the target population aspirin intake varies with factors other than drug-induced headache, which may also modulate effectiveness; hence the average natural direct effect is needed.

Descriptive indirect effects. There is no prescriptive definition of indirect effects, but the descriptive one transfers: to assess a patient's natural indirect effect, withhold treatment and ask whether the patient would recover if given as much aspirin as under treatment. (The prescriptive formulation cannot be transported, since the direct effect cannot be blocked by holding variables constant.) Their operational meaning concerns non-standard policies: if asking applicants about gender is made illegal (and no other indication of gender is available), any residual sex preference in hiring is attributable to the indirect effect of sex via qualifications. Its magnitude, predicted from data collected before the policy, is an average indirect effect.

11.1.3 Mathematisation

Motivation. Define controlled and natural direct effects and identify the latter.

Let X be the control variable, Y the response, Z the set of all intermediate variables between X and Y, and $Y_x(u)$ the value of Y for unit u under $\text{do}(X = x)$.

Definition (Controlled direct effect)

$\text{CDE}_z(x, x^*; Y, u) = Y_{xz}(u) - Y_{x^*z}(u)$, and $\text{CDE}_z(x, x^*; Y) = \mathbb{E}[Y_{xz} - Y_{x^*z}]$, the expectation taken over u.

(The difference is the most common measure; other contrasts, e.g. ratios, can be used.)

Definition (Natural direct effect)

$\text{NDE}(x, x^*; Y, u) = Y_{x,Z_{x^*}(u)}(u) - Y_{x^*}(u)$, and $\text{NDE}(x, x^*; Y) = \mathbb{E}[Y_{x,Z_{x^*}}] - \mathbb{E}[Y_{x^*}]$.

11.1.3.1 The cross-world problem

$Y_{x,Z_{x^*}}$ refers to two intervention worlds at once: world 1 with $X = x$ (to evaluate Y) and world 2 with $X = x^*$ (to determine Z_{x^*}). No experiment, not even joint randomization of X and Z, observes both worlds for the same individual. NDE and NIE are therefore empirically unverifiable in principle, and identification requires an untestable *cross-world independence* $Y_{xz} \perp\!\!\!\perp Z_{x^*}$ (Hernán and Robins 2020, §23.1; Robins and Richardson 2010).

11.1.3.2 Experimental identification

Formally, $\text{NDE}(x, x^*; Y) = \mathbb{E}[Y_{x,Z_{x^*}}] - \mathbb{E}[Y_{x^*}]$ is not in general reducible to expressions of the form $P(Y_x = y)$ or $P(Y_{xz} = y)$. (Pearl's paper is from 2001; general nested-counterfactual identification results came later: Shpitser and Pearl 2008; Correa, Lee and Bareinboim 2021.)

Theorem 11.1 (Pearl 2001)

If there is a set W of covariates, non-descendants of X or Z, with $Y_{xz} \perp\!\!\!\perp Z_{x^*} \mid W$, the average NDE is experimentally identifiable:

$$\text{NDE}(x, x^*; Y) = \sum_{w,z} \big[\mathbb{E}(Y_{xz} \mid w) - \mathbb{E}(Y_{x^*z} \mid w)\big] P(Z_{x^*} = z \mid w) P(w).$$

Proof [Supplement].

$$\mathbb{E}[Y_{x,Z_{x^*}}] = \sum_w \sum_z \mathbb{E}(Y_{xz} \mid Z_{x^*} = z, w) P(Z_{x^*} = z \mid w) P(w) = \sum_w \sum_z \mathbb{E}(Y_{xz} \mid w) P(Z_{x^*} = z \mid w) P(w),$$
$$\mathbb{E}[Y_{x^*}] = \mathbb{E}[Y_{x^*,Z_{x^*}}] = \sum_w \sum_z \mathbb{E}(Y_{x^*z} \mid w) P(Z_{x^*} = z \mid w) P(w),$$

using (in the first line) $Y_{xz} = Y_{x,Z_{x^*}}$ on the event $Z_{x^*} = z$ and the conditional independence, and (in the second) the composition axiom $Y_{x^*} = Y_{x^*,Z_{x^*}}$. Subtract. ■

Graphical criterion. Example (a): $X \to Z \to Y$, $X \to Y$, and W a common cause of Z and Y. In (b), the subgraph $\mathcal{G}_{\underline{XZ}}$ obtained by deleting all arrows emanating from X and Z, W d-separates Y from Z: $(Y \perp_d Z \mid W)_{\mathcal{G}_{\underline{XZ}}}$, which ensures the condition.

11.1.3.3 Markovian models

The condition $Y_{xz} \perp\!\!\!\perp Z_{x^*} \mid W$ holds with $W = \emptyset$ when Y_{xz} is independent of all variables in the model. In Markovian models:

- $P(Y_{xz} = y) = P(y \mid x, z)$, where $X \cup Z$ is the parent set of Y;
- $P(Z_{x^*} = z) = \sum_s P(z \mid x^*, s) P(s)$;
- $P(Y_{x,Z_{x^*}} = y) = \sum_s \sum_z P(y \mid x, z) P(z \mid x^*, s) P(s)$,

where S are the parents of Z other than X (or any set satisfying the back-door criterion). Hence the NDE is identifiable from non-experimental data:

$$\mathrm{NDE}(x, x^*; Y) = \sum_s \sum_z [\mathbb{E}(Y \mid x, z) - \mathbb{E}(Y \mid x^*, z)] P(z \mid x^*, s) P(s).$$

11.1.3.4 The mediation formula

In the simplest case ($X \to Z \to Y$, $X \to Y$, no confounding of Z–Y),

$$\mathbb{E}[Y_{x,Z_{x^*}}] = \sum_z \mathbb{E}[Y \mid X = x, Z = z]\, P(Z = z \mid X = x^*) :$$

evaluate Y under treatment x with the mediator distribution of treatment x^*. Smoking example (A treatment, M mediator): what would MI risk be if patients continued smoking ($a = 1$) but had the blood pressure of quitters ($a = 0$)?

$$\mathbb{E}[Y_{1,M_0}] = \sum_m \mathbb{E}[Y \mid A = 1, M = m]\, P(M = m \mid A = 0).$$

11.1.4 Natural indirect effects

Motivation. Define the NIE, relate it to the total effect, and identify it.

Definition (Natural indirect effect)

$\mathrm{NIE}(x, x^*; Y, u) = Y_{x^*, Z_x(u)}(u) - Y_{x^*}(u)$, and $\mathrm{NIE}(x, x^*; Y) = \mathbb{E}(Y_{x^*, Z_x}) - \mathbb{E}(Y_{x^*})$: keep the treatment at the reference level but let the mediator respond as under treatment.

Decomposition.

$$\begin{aligned} \mathrm{TE}(x, x^*; Y) &= \mathbb{E}(Y_x) - \mathbb{E}(Y_{x^*}), \\ \mathrm{NDE}(x, x^*; Y) &= \mathbb{E}(Y_{x, Z_{x^*}}) - \mathbb{E}(Y_{x^*}), \\ -\mathrm{NIE}(x^*, x; Y) &= -\mathbb{E}(Y_{x, Z_{x^*}}) + \mathbb{E}(Y_x). \end{aligned}$$

Adding the last two lines gives

$$\mathrm{TE}(x, x^*; Y) = \mathrm{NDE}(x, x^*; Y) - \mathrm{NIE}(x^*, x; Y) :$$

the total effect is the direct effect of the transition $x^* \to x$ minus the indirect effect of the reverse transition. Strange as it looks, in the linear case it reduces to the standard additive relation $\mathrm{TE}(x, x^*; Y) = \mathrm{NDE}(x, x^*; Y) + \mathrm{NIE}(x, x^*; Y)$ (Appendix 11.1.6).

Theorem 11.2 (Identification of the NIE)

If there is a set W of non-descendants of X or Z with $Y_{xz} \perp\!\!\!\perp Z_{x^*} \mid W$ for all x and z, then

$$\mathrm{NIE}(x, x^*; Y) = \sum_{w,z} \mathbb{E}(Y_{x^* z} \mid w)[P(Z_x = z \mid w) - P(Z_{x^*} = z \mid w)] P(w).$$

In non-experimental studies the NIE is identifiable when $\mathbb{E}(Y_{x^* z} \mid w)$, $P(Z_x = z \mid w)$ and $P(Z_{x^*} = z \mid w)$ are. In a simple Markovian model,

$$\mathrm{NIE}(x, x^*; Y) = \sum_z \mathbb{E}(Y \mid x^*, z)[P(z \mid x) - P(z \mid x^*)].$$

Proof [Supplement]. By definition $\mathrm{NIE}(x, x^*; Y) = \mathbb{E}(Y_{x^*,Z_x}) - \mathbb{E}(Y_{x^*}) = \mathbb{E}(Y_{x^*,Z_x}) - \mathbb{E}(Y_{x^*,Z_{x^*}})$ (composition). For the first term, condition on W and on $Z_x = z$:

$$\mathbb{E}(Y_{x^*,Z_x}) = \sum_{w,z} \mathbb{E}(Y_{x^*z} \mid Z_x = z, w) P(Z_x = z \mid w) P(w) = \sum_{w,z} \mathbb{E}(Y_{x^*z} \mid w) P(Z_x = z \mid w) P(w),$$

using $Y_{x^*z} \perp\!\!\!\perp Z_x \mid W$. The same computation with x^* in place of x gives the second term, and subtracting yields the formula. In a Markovian model with $W = \emptyset$, $\mathbb{E}(Y_{x^*z}) = \mathbb{E}(Y \mid x^*, z)$ and $P(Z_x = z) = P(z \mid x)$, which gives the last display. ■

11.1.5 Path-specific effects

Motivation. Generalize direct/indirect effects to effects along any chosen set of paths.

With several mediators ($X \to W \to Z \to Y$, $X \to Z$, $X \to Y$, ...) one may want the effect transmitted along a specific subset of paths. In the counterfactual graph, the inputs of each mechanism are switched between their values under x and under x^* according to whether the edge belongs to the selected paths.

Definition (Path-specific effect; Pearl 2001)

Let $\mathcal{G}$ be the causal graph of $\mathcal{M}$ and g an edge-subgraph of $\mathcal{G}$ containing the paths selected for analysis. The g-specific effect of x on Y (relative to the reference x^*) is the total effect of x on Y in a modified model $\mathcal{M}_g$ built as follows. Partition each parent set as $Pa_i = \{Pa_i(g), Pa_i(\bar{g})\}$, where $Pa_i(g)$ are the parents linked to X_i in g and $Pa_i(\bar{g})$ the rest. Replace each $f_i(\mathbf{pa}_i, u)$ by

$$f_i(\mathbf{pa}_i, u; g) = f_i(\mathbf{pa}_i(g), \mathbf{pa}_i^*(\bar{g}), u),$$

where $\mathbf{pa}_i^*(\bar{g}) = Pa_i(\bar{g})_{x^*}$ are the values those variables would attain (in $\mathcal{M}$ and u) under $X = x^*$. Then $SE_g(x, x^*; Y, u)_{\mathcal{M}} = TE(x, x^*; Y, u)_{\mathcal{M}_g}$.

With $g = \{X \to Y\}$ this gives the NDE; with g the mediated paths, the corresponding indirect effect. Such effects lead to nested counterfactuals such as $Y_{W_x,Z_x,Z_{x^*}}$-type expressions, represented in counterfactual graphs.

Conclusions. Pearl (2001) defines path-specific effects by *path switching* rather than variable fixing, extending the interpretation and estimation of direct and indirect effects to nonlinear models. Under certain conditions these effects are identifiable even when Z is affected by X; the conditions always hold in Markovian models. The definition also gives an operational interpretation of indirect effects.

11.1.6 Appendix: The linear additive relation

Take $Z = \alpha X + U_Z$, $Y = \beta X + \gamma Z + U_Y$, so $Z_x = \alpha x + U_Z$ and $Z_{x^*} = \alpha x^* + U_Z$. Then

$$\begin{aligned}
\mathrm{TE}(x, x^*; Y) &= \mathbb{E}(Y_x) - \mathbb{E}(Y_{x^*}) = (\beta + \alpha\gamma)(x - x^*), \\
\mathrm{NDE}(x, x^*; Y) &= \mathbb{E}(Y_{x,Z_{x^*}}) - \mathbb{E}(Y_{x^*}) = \beta(x - x^*), \\
\mathrm{NIE}(x, x^*; Y) &= \mathbb{E}(Y_{x^*,Z_x}) - \mathbb{E}(Y_{x^*}) = \alpha\gamma(x - x^*),
\end{aligned}$$

so TE = NDE + NIE. (Also $\mathrm{NIE}(x^*, x; Y) = -\alpha\gamma(x - x^*)$, consistent with $\mathrm{TE} = \mathrm{NDE}(x, x^*) - \mathrm{NIE}(x^*, x)$.) This uses the linear, no-interaction form; nonlinear models do not generally preserve the same-direction sum.

11.1.7 Appendix: Counterfactual axioms

Composition. For singleton Y, W and any set X: $W_x(u) = w \Rightarrow Y_{xw}(u) = Y_x(u)$. Forcing a variable to the value it would have had anyway has no effect on other variables.
Consistency. $X(u) = x \Rightarrow Y(u) = Y_x(u)$.
Effectiveness. For all X, W: $X_{xw}(u) = x$.
Reversibility. For Y, W and any X: $(Y_{xw}(u) = y) \wedge (W_{xy}(u) = w) \Rightarrow Y_x(u) = y$.

11.2 Single-world intervention graphs

Motivation. The central question is how potential outcomes can be placed on a causal graph, so that counterfactual independences can be read off by d-separation, without the extra untestable assumptions of a fully specified structural-equation model.

11.2.1 Potential outcomes and graphs

Motivation. Recall the two languages that SWIGs unify.

11.2.1.1 Potential outcomes

For binary treatment X and response Y: $Y(x = 0)$ (also $Y(x_0)$) is the value of Y that would be observed for a unit assigned $X = 0$; $Y(x = 1)$ (or $Y(x_1)$) likewise for $X = 1$. This presumes the outcomes are well defined: one version of treatment, and no interference between units (SUTVA). "Potential outcome" and "counterfactual" are used synonymously.

Response types for binary Y:

$Y(x_0)$	$Y(x_1)$	Name
0	0	never recover
0	1	helped
1	0	hurt
1	1	always recover

Relating counterfactuals and do-notation. $P(Y(x) = y) \equiv P(Y = y \mid \mathrm{do}(X = x))$, but counterfactual notation is more general. Example: the distribution of outcomes that would have arisen among those who took treatment $(X = 1)$ had they, counter to fact, not received it: $P(Y(x = 0) = y \mid X = 1)$. If treatment is randomized, $X \perp\!\!\!\perp Y(x = 0)$ and this equals $P(Y(x = 0) = y)$; in an observational study they may differ.

11.2.1.2 Graphs

A DAG is associated with the factorization $P(\mathbf{V}) = \prod_{X \in \mathbf{V}} P(X \mid \mathrm{pa}(X))$; e.g. for $A \to B$, $A \to C$, $B \to D$, $C \to D$, $D \to E$, $P(A, B, C, D, E) = P(A)P(B \mid A)P(C \mid A)P(D \mid B, C)P(E \mid D)$. d-separation reads independences such as $C \perp\!\!\!\perp B \mid A$, $D \perp\!\!\!\perp A \mid B, C$, $E \perp\!\!\!\perp A, B, C \mid D$.

In the graphical approach, edges represent direct causal relations, and by convention confounders (e.g. an unobserved H with $X \leftarrow H \to Y$) are included. It originates in Sewall Wright's path diagrams of the 1920s. For $X \to Y$ without confounding, $P(x, y) = P(x)P(y \mid x)$ and the causal model asserts

$$P(Y(x) = y) = P(Y = y \mid \mathrm{do}(X = x)) = P(Y = y \mid X = x),$$

where $P(y \mid \text{do}(x))$ is the distribution under an experiment fixing X to x. How does this relate to the counterfactual approach? No confounding corresponds to $X \perp\!\!\!\perp Y(x_0)$ and $X \perp\!\!\!\perp Y(x_1)$; confounding to $X \not\perp\!\!\!\perp Y(x_0)$ or $X \not\perp\!\!\!\perp Y(x_1)$. The elephant in the room: $Y(x_0)$ and $Y(x_1)$ do not appear on these graphs.

11.2.2 Node splitting

Motivation. Put counterfactual variables on the graph.

Setting X to 0. Start from $P(X = \tilde{x}, Y = \tilde{y}) = P(X = \tilde{x})P(Y = \tilde{y} \mid X = \tilde{x})$ for $X \to Y$. Split X into a random node X and a fixed node $x = 0$; the outcome becomes $Y(x = 0)$:



Now $X \perp\!\!\!\perp Y(x = 0)$ can be read off, and a new factorization is associated:

$$P(X = \tilde{x}, Y(x = 0) = \tilde{y}) = P(X = \tilde{x})\, P(Y(x = 0) = \tilde{y}), \qquad P(Y(x = 0) = \tilde{y}) = P(Y = \tilde{y} \mid X = 0).$$

The last equation links a term of the original factorization to the new one: the *modularity assumption*.

Marginals represented by SWIGs are identified. The SWIG $\mathcal{G}(x_0)$ represents $P(X, Y(x_0))$ and $\mathcal{G}(x_1)$ represents $P(X, Y(x_1))$; under no confounding both are identified from $P(X, Y)$. In contrast $P(X, Y(x_0), Y(x_1))$ is not identified. $Y(x = 0)$ and $Y(x = 1)$ never appear on the same graph: although $X \perp\!\!\!\perp Y(x = 0)$ and $X \perp\!\!\!\perp Y(x = 1)$, we do not assume $X \perp\!\!\!\perp \{Y(x = 0), Y(x = 1)\}$. A single graph containing both would have been impossible; hence *single-world* intervention graphs.

Templates. Both graphs are represented by a template $\mathcal{G}(x)$: $X \mid x \to Y(x)$. Formally the template is a graph-valued function (not a graph): it takes a value x and returns a SWIG $\mathcal{G}(x)$; each instantiation represents a different margin.

Intuition (Robins, VanderWeele and Richardson 2007). How could we learn whether someone would choose treatment $(X = 1)$ and, at the same time, what happens to that person without treatment $(Y(x = 0))$? Imagine that whenever a patient is observed to swallow the drug $(X = 1)$, we instantly administer a safe emetic that makes them regurgitate the pill before any drug enters the bloodstream. With no side effects of the emetic, the recorded outcome is $Y(x = 0)$, observed jointly with the natural choice X.

11.2.3 SWIG construction and properties

Motivation. Construct SWIGs in general and state what they imply.

Given $\mathcal{G}$ and a set $\mathbf{A} = \{A_1, \ldots, A_k\}$ to intervene on, form $\mathcal{G}(\mathbf{a})$ in two steps:

1. **Node splitting:** split every $A \in \mathbf{A}$ into a random node A and a fixed node a. The random half inherits all edges directed *into* A; the fixed half inherits all edges directed *out of* A.
2. **Relabelling:** every descendant Y of a fixed node is relabelled $Y(\mathbf{a}_Y)$, indexed by the fixed nodes that are its ancestors (e.g. $B(a, \ldots)$, $C(a, \ldots)$, $D(a, \ldots)$); non-descendants keep their names.

Factorization and modularity. The original graph $\mathcal{G}$ carries the observed $P(\mathbf{V})$; the SWIG $\mathcal{G}(\tilde{\mathbf{a}})$ carries the counterfactual distribution $P(\mathbf{V}(\tilde{\mathbf{a}}))$.

- *Factorization:* $P(\mathbf{V}(\tilde{\mathbf{a}}))$ factorizes with respect to $\mathcal{G}(\tilde{\mathbf{a}})$, ignoring fixed nodes.

- *Modularity:* the conditional density of $Y(\tilde{\mathbf{a}}_Y)$ in $\mathcal{G}(\tilde{\mathbf{a}})$ equals that of Y in $\mathcal{G}$ after substituting $\tilde{a}_i$ for any $A_i \in \mathbf{A}$ that is a parent of Y.

Consequence: if $P(\mathbf{V})$ is observed, $P(\mathbf{V}(\tilde{\mathbf{a}}))$ is identified.

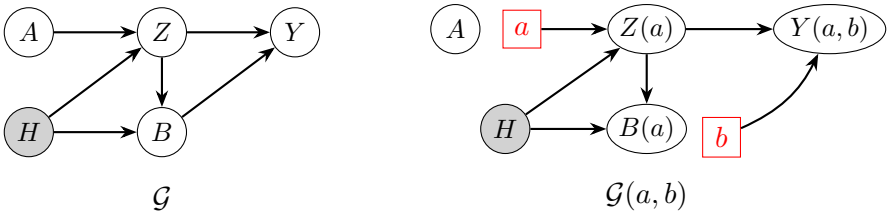
d-separation on SWIGs. If subsets $\mathbf{B}(\tilde{\mathbf{a}})$ and $\mathbf{C}(\tilde{\mathbf{a}})$ of random nodes are d-separated by $\mathbf{D}(\tilde{\mathbf{a}})$ together with the fixed nodes $\tilde{\mathbf{a}}$ in $\mathcal{G}(\tilde{\mathbf{a}})$, then

$$\mathbf{B}(\tilde{\mathbf{a}}) \perp\!\!\!\perp \mathbf{C}(\tilde{\mathbf{a}}) \mid \mathbf{D}(\tilde{\mathbf{a}}) \quad \text{in } P(\mathbf{V}(\tilde{\mathbf{a}})).$$

11.2.4 A harder inferential problem

Motivation. Use a SWIG to settle a counterfactual independence query.

Graph $\mathcal{G}$: $A \to Z$, $Z \to Y$, $Z \to B$, $B \to Y$, and an unobserved H with $H \to Z$, $H \to B$. Query: does $\mathcal{G}$ imply $Y(a,b) \perp\!\!\!\perp B(a) \mid Z(a), A$?



$\mathcal{G}$ $\qquad$ $\mathcal{G}(a,b)$

Pearl (2009, Ex. 11.3.3) claims the DAG does not imply

$$Y(a,b) \perp\!\!\!\perp B \mid Z, A = a. \tag{3}$$

The SWIG shows that (3) does hold. In $\mathcal{G}(a,b)$, every path from $B(a)$ to $Y(a,b)$ passes through $Z(a)$ as a non-collider ($B(a) \leftarrow Z(a) \to Y(a,b)$, $B(a) \leftarrow H \to Z(a) \to Y(a,b)$), and $B(a)$ has no edge to $Y(a,b)$ because the edge $B \to Y$ now leaves the fixed node b. Hence

$$Y(a,b) \perp\!\!\!\perp B(a) \mid Z(a), A \quad \Longrightarrow \quad Y(a,b) \perp\!\!\!\perp B(a) \mid Z(a), A = a,$$

which is equivalent to (3) by consistency ($B(a) = B$ and $Z(a) = Z$ when $A = a$). (Pearl had inferred that a claim of Robins was false because, if true, (3) would hold.)

11.2.5 Adjustment for confounding

Motivation. Prove the back-door formula in the SWIG language.

Graph: $L \to X$, $L \to Y$, $X \to Y$. In the SWIG $\mathcal{G}(x)$: $L \to X$, $L \to Y(x)$, $x \to Y(x)$, so $X \perp\!\!\!\perp Y(x) \mid L$. Then

$$\begin{aligned} P[Y(\tilde{x}) = y] &= \sum_l P[Y(\tilde{x}) = y \mid L = l] P(L = l) \\ &= \sum_l P[Y(\tilde{x}) = y \mid L = l, X = \tilde{x}] P(L = l) && (X \perp\!\!\!\perp Y(\tilde{x}) \mid L) \\ &= \sum_l P[Y = y \mid L = l, X = \tilde{x}] P(L = l) && \text{(consistency)}. \end{aligned}$$

Connection to do-calculus. Factorization and modularity suffice to imply all identification results of Pearl's (1995) do-calculus (see also Spirtes et al. 1993): $P(Y = y \mid \text{do}(A = a))$ is identified iff $P(Y(a) = y)$ is identified.

11.2.6 Conclusion

Eliminating a false trichotomy. Previously the only unification of counterfactuals and graphs was Pearl's non-parametric structural equation models with independent errors, which left three options: use graphs without counterfactuals (Dawid); use counterfactuals without graphs (many statisticians); or use both but accept many additional assumptions that are not experimentally testable even in principle and not needed for most identification results. SWIGs allow graphs and counterfactuals together without these extra assumptions.

Summary and extensions. SWIGs unify graphs and counterfactuals via node splitting, working by linking the factorizations of the two graphs. The new graph represents a counterfactual distribution identified from the distribution of the original DAG, giving a common language for counterfactual and graphical researchers. (Not covered: combining information on the absence of individual- and population-level direct effects; models in which interventions on only some variables are well defined.)

References: Pearl (2009); Richardson and Robins (2013), CSSS Technical Report 128; Robins (1986); Robins, VanderWeele and Richardson (2007); Shpitser and Pearl (2008); Spirtes, Glymour and Scheines (1993).

11.3 G-methods for time-varying treatments

Motivation. The central question is how to estimate the effect of a sequence of treatment decisions when covariates measured along the way are both affected by earlier treatment and drive later treatment, a situation in which ordinary regression, stratification and matching all fail.

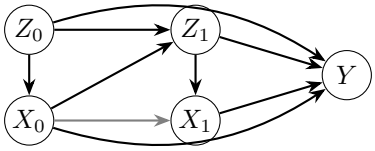
11.3.1 Time-varying treatments

Motivation. Set up sequential decisions and explain why feedback breaks standard adjustment.

Recall the single time point. Identification by back-door adjustment, $\mathbb{E}[Y(x)] = \sum_z \mathbb{E}[Y \mid X = x, Z = z]P(Z = z)$, and estimation by IPW, $\mathbb{E}[Y(x)] = \mathbb{E}[\mathbb{1}(X = x)Y/P(X = x \mid Z)]$. What if treatment is applied repeatedly?

Many interventions involve repeated decisions: medication updated at each visit, policies applied in stages, algorithms adapting to the current state. Later decisions are made after earlier information has been observed, so the data-generating process is history-dependent: action $\to$ state $\to$ action $\to \cdots \to$ outcome.

Causal diagrams. Treatments X_k with history $\bar{X}_k = (X_0, \ldots, X_k)$; covariates (states) Z_k with history $\bar{Z}_k = (Z_0, \ldots, Z_k)$; outcome Y at the end of follow-up. For two periods: $Z_0 \to X_0$, $Z_0 \to Z_1$, $X_0 \to Z_1$, $Z_1 \to X_1$, $X_0 \to X_1$, and all of them $\to Y$.



Treatment–confounder feedback occurs when (1) past treatment X_{k-1} affects a future confounder Z_k, and (2) Z_k affects future treatment X_k. Example (HIV): antiretroviral therapy (ART, X) $\to$ CD4 count or viral load (Z, health biomarkers) $\to$ next ART decision $\to \cdots$. Previous treatment may also guide later decisions directly ($X_{k-1} \to X_k$). This feedback loop makes standard methods fail.

Treatment strategies. A strategy specifies treatment at each time. *Static*: the same for everyone regardless of covariates, e.g. "always treat" $\bar{x} = (1, \ldots, 1)$, "never treat" $\bar{x} = (0, \ldots, 0)$. *Dynamic*: treatment at time k depends on observed history, $g = [g_0(Z_0), g_1(\bar{X}_0, Z_1), \ldots, g_K(\bar{X}_{K-1}, Z_K)]$, e.g. "initiate ART only when CD4 drops below 500". Causal estimands contrast mean counterfactual outcomes, $\mathbb{E}[Y(\bar{x})] - \mathbb{E}[Y(\bar{x}')]$.

11.3.1.1 Identification assumptions

Sequentially randomized experiments: X_k is randomly assigned with probability depending only on $(\bar{X}_{k-1}, \bar{Z}_k)$. The observational analogue: no unmeasured common causes of X_k and Y given $(\bar{X}_{k-1}, \bar{Z}_k)$. An unmeasured U may affect Y and Z_1 as long as it has no arrow into X_0 or X_1; then adjusting for the measured past suffices.

- **Sequential exchangeability:** $Y(\bar{x}) \perp\!\!\!\perp X_k \mid \bar{X}_{k-1}, \bar{Z}_k$ for $k = 0, \ldots, K$ (treatment as-if randomized given the observed past).
- **Consistency:** $Y(\bar{x}) = Y$ when $\bar{X} = \bar{x}$.
- **Positivity:** $P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k) > 0$.

11.3.1.2 The fundamental dilemma

Should we adjust for Z_1? Adjusting removes confounding of $X_1 \to Y$ (via Z_1) but over-adjusts by blocking $X_0 \to Z_1 \to Y$ (and, with $U \to Z_1$, $U \to Y$, opens the collider $X_0 \to Z_1 \leftarrow U \to Y$). Not adjusting preserves X_0's effect through Z_1 but leaves X_1 confounded. Traditional methods (regression, stratification, matching) cannot do both under feedback.

G-methods (Robins' generalized methods) are consistent under time-varying confounding:

(1) the g-formula (outcome modeling);

(2) IP-weighted marginal structural models (treatment modeling);

(3) g-estimation of structural nested models (effect modeling). All rely on the same assumptions. Key idea: handle Z_1 without blocking the causal path through it.

What you already know. Apply the truncated factorization (Chapter 2) to (Z_0, X_0, Z_1, X_1, Y): from

$$P(z_0) \cdot P(x_0 \mid z_0) \cdot P(z_1 \mid x_0, z_0) \cdot P(x_1 \mid z_1, x_0, z_0) \cdot P(y \mid x_1, z_1, x_0, z_0)$$

remove the treatment factors ($\text{do}(x_0)$, $\text{do}(x_1)$) and keep the rest:

$$\mathbb{E}[Y(x_0, x_1)] = \sum_{z_0, z_1} \mathbb{E}[Y \mid x_0, z_0, x_1, z_1]\, P(z_1 \mid x_0, z_0)\, P(z_0).$$

This is the *g-formula*, the truncated factorization in temporal order; sequential exchangeability ensures each remaining factor is estimable. What is new: why feedback breaks standard estimation, and estimation strategies (sequential weight products, MSMs, SNMs).

11.3.2 The g-formula

Motivation. Standardize over the time-varying confounder instead of conditioning on it.

Idea (two time points, no baseline confounder, so X_0 is marginally exchangeable). Instead of conditioning on Z_1, standardize over its distribution *given past treatment*:

$$\mathbb{E}[Y(x_0, x_1)] = \sum_{z_1} \mathbb{E}[Y \mid X_0 = x_0, Z_1 = z_1, X_1 = x_1]\, P(Z_1 = z_1 \mid X_0 = x_0),$$

weighting by $P(z_1 \mid x_0)$, not by the marginal of Z_1.

Derivation. The observed density factorizes temporally (writing f and P interchangeably):

$$P(y, x_1, z_1, x_0) = P(y \mid x_1, z_1, x_0) \underbrace{P(x_1 \mid z_1, x_0)}_{\text{remove}} P(z_1 \mid x_0) \underbrace{P(x_0)}_{\text{remove}}.$$

Under $\mathrm{do}(X_0 = x_0, X_1 = x_1)$ the assignment terms are replaced by the intervention, giving outcome model $\times$ confounder model as above. For $K+1$ time points,

$$\mathbb{E}[Y(\bar{x})] = \sum_{\bar{z}} \mathbb{E}[Y \mid \bar{X} = \bar{x}, \bar{Z} = \bar{z}] \prod_{k=0}^{K} P(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1}),$$

with $\bar{x}_{-1}, \bar{z}_{-1}$ empty so that $P(z_0 \mid \bar{x}_{-1}, \bar{z}_{-1}) = P(z_0)$.

11.3.2.1 Iterated conditional expectations (ICE)

The sum over $\bar{z}$ needs covariate densities $P(z_k \mid \text{past})$, hard to model for high-dimensional Z_k. ICE replaces the sum by nested regressions computed backward in time: each step is an outcome regression averaging out one confounder, the data doing the averaging, so no density model is needed:

$$Y \xrightarrow{\text{regress}} \hat{Q}_K \to \cdots \to \hat{Q}_0 \xrightarrow{\text{average}} \hat{\mathbb{E}}[Y(\bar{x})].$$

Two time points. (1) $\hat{Q}_1(z_1) = \hat{\mathbb{E}}[Y \mid x_0, z_1, x_1]$, an outcome regression at (x_0, x_1);

(2) $\hat{\mathbb{E}}[Y(x_0, x_1)] = \hat{\mathbb{E}}[\hat{Q}_1(Z_1) \mid X_0 = x_0]$, the average over the $X_0 = x_0$ group. The second step performs the g-formula's averaging through the empirical distribution of Z_1 in the selected treatment-history group.

Three time points as variable elimination. Eliminate Z_2, then Z_1:

$$Q_2(z_1, z_2) = \mathbb{E}[Y \mid x_0, z_1, x_1, z_2, x_2], \qquad Q_1(z_1) = \mathbb{E}[Q_2(Z_1, Z_2) \mid x_0, z_1, x_1],$$
$$\mathbb{E}[Y(x_0, x_1, x_2)] = \mathbb{E}[Q_1(Z_1) \mid X_0 = x_0].$$

For Q_1, create the pseudo-outcome $\tilde{Y}_i^{(2)} = \hat{Q}_2(Z_{1i}, Z_{2i})$ with treatments set to (x_0, x_1, x_2), and regress it on Z_1 among rows with $X_0 = x_0$, $X_1 = x_1$; the data distribution averages over Z_2.

General recursion. Starting from $\hat{Q}_{K+1} = Y$, for $k = K, \ldots, 0$:

$$\hat{Q}_k = \hat{\mathbb{E}}[\hat{Q}_{k+1} \mid \bar{X}_k = \bar{x}_k, \bar{Z}_k], \qquad \hat{\mathbb{E}}[Y(\bar{x})] = \hat{\mathbb{E}}[\hat{Q}_0(x_0, Z_0)].$$

Every step is an outcome regression; ICE never models $P(z_k \mid \text{past})$, which is why it scales to many time points and high-dimensional Z_k.

11.3.3 IP weighting and marginal structural models

Motivation. Re-weight to a pseudo-population in which treatment is unconfounded.

Idea. Instead of modeling the outcome, weight each individual inversely to the probability of their observed treatment given confounders; this "breaks" the arrows $Z_k \to X_k$ (and U, which never points into treatment, stays harmless).

One time point: why $1/P(X \mid Z)$. Under exchangeability $Y(x) \perp\!\!\!\perp X \mid Z$, consistency and positivity,

$$\mathbb{E}\Big[\frac{\mathbb{1}(X = x)Y}{P(x \mid Z)}\Big] = \mathbb{E}_Z\Big[\frac{\mathbb{E}[\mathbb{1}(X = x)Y \mid Z]}{P(x \mid Z)}\Big] = \mathbb{E}_Z\Big[\frac{\mathbb{E}[Y \mid X = x, Z]P(x \mid Z)}{P(x \mid Z)}\Big]$$
$$= \mathbb{E}_Z\big[\mathbb{E}[Y \mid X = x, Z]\big] = \mathbb{E}_Z\big[\mathbb{E}[Y(x) \mid Z]\big] = \mathbb{E}[Y(x)].$$

Dividing by $P(x \mid Z)$ up-weights under-represented (X, Z) combinations, building a pseudo-population with $X \perp\!\!\!\perp Z$: the arrow $Z \to X$ is gone and the crude association becomes causal.

Two time points (no baseline confounder), for a fixed regime (x_0, x_1):

$$W^X(x_0, x_1) = \frac{\mathbb{1}(X_0 = x_0)}{P(x_0)} \cdot \frac{\mathbb{1}(X_1 = x_1)}{P(x_1 \mid x_0, Z_1)},$$

$$\begin{aligned}\mathbb{E}[W^X(x_0, x_1)Y] &= \sum_{z_1} \mathbb{E}[Y \mid x_0, z_1, x_1] \frac{P(x_0, z_1, x_1)}{P(x_0)P(x_1 \mid x_0, z_1)} \\ &= \sum_{z_1} \mathbb{E}[Y \mid x_0, z_1, x_1] P(z_1 \mid x_0) = \mathbb{E}[Y(x_0, x_1)].\end{aligned}$$

The first factor undoes the observed assignment of X_0, the second that of X_1 given history (X_0, Z_1). With baseline Z_0, replace $P(X_0 = x_0)$ by $P(X_0 = x_0 \mid Z_0)$.

The weight product. Confounding $Z_k \to X_k$ enters at every step, so divide by $P(X_k \mid \text{past})$ at every k:

$$W^X = \prod_{k=0}^{K} \frac{1}{P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)} \qquad \text{(non-stabilized)}.$$

Stabilized weights. The mean of W^X blows up: for binary treatments,

$$\mathbb{E}[W^X] = \sum_{\bar{x}, \bar{z}} \frac{P(\bar{x}, \bar{z})}{\prod_k P(x_k \mid \text{past})} = \sum_{\bar{x}, \bar{z}} \prod_k P(z_k \mid \text{past}) = \sum_{\bar{x}} 1 = 2^{K+1},$$

so the pseudo-population is 2^{K+1} times larger, and rare treatment cells give extreme weights and high variance. Fix: put $P(X_k \mid \bar{X}_{k-1})$ (treatment given past treatment only) in the numerator,

$$SW^X = \prod_{k=0}^{K} \frac{P(X_k \mid \bar{X}_{k-1})}{P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)}.$$

The numerator uses treatment history only, so it does not reintroduce $Z \to X$ confounding (the estimator remains valid), and $\mathbb{E}[SW^X] = 1$: the pseudo-population has the data's size, with weights centered at 1 and much smaller variance.

Proof (one time point). With $SW = P(X)/P(X \mid Z)$,

$$\begin{aligned}\mathbb{E}\Big[\frac{P(x)}{P(x \mid Z)} \mathbb{1}(X = x) Y\Big] &= P(x)\, \mathbb{E}_Z\Big[\frac{\mathbb{E}[Y \mid X = x, Z] P(x \mid Z)}{P(x \mid Z)}\Big] \\ &= P(x)\, \mathbb{E}_Z\big[\mathbb{E}[Y \mid X = x, Z]\big] = P(x)\, \mathbb{E}[Y(x)].\end{aligned}$$

With $Y \equiv 1$, $\mathbb{E}[SW\, \mathbb{1}(X = x)] = P(x)$, so $\mathbb{E}[SW] = \sum_x P(x) = 1$ and $\mathbb{E}[SW\, \mathbb{1}(X = x)Y]/\mathbb{E}[SW\, \mathbb{1}(X = x)] = \mathbb{E}[Y(x)]$. The extra factor $P(x)$ merely rescales the $X = x$ arm to its marginal size, so the pseudo-population is not doubled; it cancels in the ratio (and in an MSM), leaving the causal target unchanged and only shrinking variance.

Proof (general). Let $r_k(x_k \mid \bar{x}_{k-1}) \equiv P(X_k = x_k \mid \bar{X}_{k-1} = \bar{x}_{k-1})$. The observed law factorizes as $p(\bar{z}, \bar{x}, y) = \prod_{k=0}^{K} p(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1}) p(x_k \mid \bar{x}_{k-1}, \bar{z}_k) \cdot p(y \mid \bar{x}, \bar{z})$. For $SW^X = \prod_k r_k(X_k \mid \bar{X}_{k-1})/p(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)$, the weighted population has density

$$p_{SW}(\bar{z}, \bar{x}, y) = p(\bar{z}, \bar{x}, y)\, SW^X = \prod_{k=0}^{K} p(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1})\, r_k(x_k \mid \bar{x}_{k-1}) \cdot p(y \mid \bar{x}, \bar{z}),$$

a proper density (so $\mathbb{E}[SW^X] = 1$) in which treatment depends only on past treatment: the $Z_k \to X_k$ arrows are removed. For a static regime $\bar{x}$ let $r(\bar{x}) = \prod_k r_k(x_k \mid \bar{x}_{k-1})$. Then

$$\mathbb{E}\big[SW^X \mathbb{1}(\bar{X} = \bar{x}) Y\big] = \sum_{\bar{z}} \mathbb{E}[Y \mid \bar{x}, \bar{z}] \prod_k p(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1})\, r(\bar{x}) = r(\bar{x})\, \mathbb{E}[Y(\bar{x})]$$

by the g-formula, and $\mathbb{E}[SW^X \not\Vdash(\bar{X} = \bar{x})] = r(\bar{x})$, so $\mathbb{E}[SW^X \not\Vdash(\bar{X} = \bar{x})Y]/\mathbb{E}[SW^X \not\Vdash(\bar{X} = \bar{x})] = \mathbb{E}[Y(\bar{x})]$. Stabilization changes the marginal treatment distribution in the pseudo-population, not the causal target.

11.3.3.1 Marginal structural models

An MSM specifies the relation between treatment strategy and counterfactual mean, e.g.

$$\mathbb{E}[Y(x_0, x_1)] = \beta_0 + \beta_1 x_0 + \beta_2 x_1 + \beta_3 x_0 x_1.$$

It is a model for potential outcomes, not for observed data, estimated by weighted regression in the IP-weighted pseudo-population. More complex MSMs: saturated (one parameter per strategy); dose–response $\mathbb{E}[Y(\bar{x})] = \beta_0 + \beta_1 \text{cum}(\bar{x})$ with $\text{cum}(\bar{x}) = \sum_k x_k$; conditional MSMs including baseline covariates.

11.3.4 G-estimation of structural nested models

Motivation. Model and remove the history-specific effect of each decision.

The blip. An MSM models $\mathbb{E}[Y(\bar{x})]$ marginally (averaged over history). A structural nested mean model (SNMM) targets the local, history-specific effect of each decision, the *blip*: a contrast of potential outcomes with the future held at the reference (0):

$$\gamma_1(x_0, z_1) = \mathbb{E}[Y(x_0, 1) - Y(x_0, 0) \mid X_0 = x_0, Z_1 = z_1] \text{ (time-1 blip, given history)},$$

$$\gamma_0(x_0) = \mathbb{E}[Y(x_0, 0) - Y(0, 0)] \text{ (time-0 blip)}.$$

Blips accumulate: in a three-decision example from "never" (000) to "always" (111), $\mathbb{E}[Y(\bar{x})]$ climbs $000 \to 100 \to 110 \to 111$ by steps $+\gamma_0, +\gamma_1, +\gamma_2$; the MSM is the whole climb, the SNMM the history-specific steps.

G-estimation: find the blip by removing it. If the late blip were known, subtracting it would rebuild each patient's outcome had they not taken the late treatment: $Y_{\text{removed}} = Y - \psi_1 X_1$ (a candidate blip). G-estimation solves for the ψ_1 that makes Y_{removed} balanced across X_1 given history: an estimating equation, not an outcome regression. Two handles, one target: IPW models the treatment (re-weighting to break confounding); SNMM models the effect (the blip) and removes it. Both recover $\mathbb{E}[Y(\bar{x})]$.

In practice: one time point. With an additive, history-homogeneous blip $\gamma(z; \psi) = \psi x$, blipping down gives $H(\psi) = Y - \psi X$, the candidate $Y(0)$. At the true ψ, consistency and the blip make $H(\psi) = Y(0)$, and exchangeability gives $Y(0) \perp\!\!\!\perp X \mid Z$, so $H(\psi)$ must be uncorrelated with X given Z. With a fitted propensity $\hat{e}(Z) = \hat{P}(X = 1 \mid Z)$, solve

$$U(\psi) = \sum_i (X_i - \hat{e}(Z_i)) \underbrace{(Y_i - \psi X_i)}_{H_i(\psi)} = 0 \quad \Longrightarrow \quad \hat{\psi} = \frac{\sum_i (X_i - \hat{e}_i) Y_i}{\sum_i (X_i - \hat{e}_i) X_i},$$

an IV-type ratio in which the propensity residual $X - \hat{e}$ plays the role of an instrument. Recipe:

(1) fit $\hat{e}(Z)$ (e.g. logistic);

(2) form residuals $X_i - \hat{e}_i$;

(3) plug in. It is consistent if $\hat{e}$ is correct; adding an outcome term $\mathbb{E}[H(\psi) \mid Z]$ makes it doubly robust. With effect modification, e.g. $\gamma(z; \psi) = (\psi_0 + \psi_1 z)x$, H uses the z-specific blip and the instrument becomes the vector $(1, z)(X - \hat{e})$.

Two time points. Order $X_0 \to Z_1 \to X_1 \to Y$ with history-homogeneous blips $\gamma_1 = \psi_1 x_1$, $\gamma_0 = \psi_0 x_0$ (history enters only through the propensities). Fit the latest blip first, then work backward.

1. Remove the time-1 blip: $H_1(\psi_1) = Y - \psi_1 X_1$ is the candidate $Y(x_0, 0)$, and exchangeability gives $Y(x_0, 0) \perp\!\!\!\perp X_1 \mid (X_0, Z_1)$. With $\hat{e}_1 = \hat{P}(X_1 = 1 \mid X_0, Z_1)$, solving $\sum_i (X_{1i} - \hat{e}_{1i}) H_{1i}(\psi_1) = 0$ gives
$$\hat{\psi}_1 = \frac{\sum_i (X_{1i} - \hat{e}_{1i}) Y_i}{\sum_i (X_{1i} - \hat{e}_{1i}) X_{1i}}.$$

2. Remove the time-0 blip from the blipped-down outcome: $H_0 = H_1(\hat{\psi}_1) - \psi_0 X_0 = Y - \hat{\psi}_1 X_1 - \psi_0 X_0$, the candidate $Y(0, 0)$. With $\hat{e}_0 = \hat{P}(X_0 = 1 \mid Z_0)$,
$$\hat{\psi}_0 = \frac{\sum_i (X_{0i} - \hat{e}_{0i}) H_{1i}(\hat{\psi}_1)}{\sum_i (X_{0i} - \hat{e}_{0i}) X_{0i}}.$$

The same one-step equation is applied backward with history-indexed propensity residuals $X_k - \hat{e}_k$. Telescoping, $\mathbb{E}[Y(1,1)] - \mathbb{E}[Y(0,0)] = \hat{\psi}_0 + \hat{\psi}_1$. Each step becomes doubly robust by adding an outcome term $\mathbb{E}[H_k \mid \text{history}]$.

11.3.5 Doubly robust estimation

Motivation. Combine outcome and treatment models for a second chance at consistency.

The g-formula needs a correct outcome model $\mathbb{E}[Y \mid \bar{X}, \bar{Z}]$; IP weighting needs correct treatment models $P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)$; misspecification biases either. Doubly robust (augmented IPW, AIPW) estimators are consistent if either model is correct.

Time-fixed review. With outcome model $\hat{m}(x, z) = \hat{\mathbb{E}}[Y \mid X = x, Z = z]$ and propensity $\hat{\pi}(z) = \hat{P}(X = 1 \mid Z = z)$,
$$\hat{\tau}_{DR} = \frac{1}{n} \sum_{i=1}^{n} \left[\hat{m}(1, Z_i) - \hat{m}(0, Z_i) + \frac{X_i \{Y_i - \hat{m}(1, Z_i)\}}{\hat{\pi}(Z_i)} - \frac{(1 - X_i)\{Y_i - \hat{m}(0, Z_i)\}}{1 - \hat{\pi}(Z_i)} \right].$$

If $\hat{m}$ is correct, the IPW residual terms have mean zero; if $\hat{\pi}$ is correct, the residuals correct the bias of $\hat{m}$.

Time-varying extension. Fit treatment models $\hat{\pi}_k$ at each time and outcome models backward from K to 0 (ICE-style); the estimator augments IPW with outcome predictions at each step and is consistent if, at each time point, either the treatment or the outcome model is correct. The full sequential AIPW involves iterated regressions and cumulative weights (Bang and Robins 2005). Modern variants: TMLE, sequential regression with cross-fitting.

11.3.6 Practical considerations

Motivation. Dynamic strategies and positivity.

Dynamic strategies. All three g-methods generalize to $g = [g_0(Z_0), g_1(\bar{X}_0, Z_1), \ldots, g_K(\bar{X}_{K-1}, Z_K)]$ with $g_k(\cdot) \in \{0, 1\}$. G-formula: simulate forward, drawing Z_k and setting $X_k = g_k(\cdot)$. IPW: weight individuals who happened to follow g,
$$W^g = \prod_{k=0}^{K} \frac{\mathbb{1}(X_k = g_k(\bar{X}_{k-1}, Z_k))}{P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)}.$$

G-estimation: SNMMs naturally accommodate dynamic regimes via effect modification by Z_k.

Positivity violations. With K binary treatment times there are 2^K treatment histories $(2, 4, 8, \ldots)$, so cells become sparse (near-positivity violations), IP weights extreme, and estimates unstable. Remedies: stabilized weights, weight truncation, parametric models.

11.3.7 Comparison of g-methods

Motivation. Summarize what each method models, targets and risks.

	G-formula	IPW / MSM	G-estimation / SNM
Key model	outcome + confounder dist.	treatment assignment	treatment + blip function
Target	$\mathbb{E}[Y(\bar{x})]$ (marginal)	$\mathbb{E}[Y(\bar{x})]$ (marginal)	conditional blip effects
Effect modification by Z_k	outcome-model interactions	not directly	directly modeled
Extreme weights?	no	yes	no
G-null paradox?	yes (parametric)	no	no
DR version?	yes	yes (AIPW)	yes

All three are consistent under correct specification and the same identifiability conditions. (The g-null paradox: with parametric models for both outcome and time-varying confounders, the parametric g-formula can reject the null of no effect even when it is true, because the models cannot all be correct under the null.)

Estimand, model, estimator.

Method	Estimand (what)	Model (what must be right)	Estimator (how)
g-formula	$\mathbb{E}[Y(\bar{x})]$	outcome (+ confounders)	standardize / ICE
IPW	$\mathbb{E}[Y(\bar{x})]$	treatment	weighting
MSM	$\mathbb{E}[Y(\bar{x})]$, marginal	*marginal form* + treatment	weighted regression
SNMM / g-est.	blip γ_k (history-specific)	*blip form* + treatment	g-estimation (balance)
AIPW / DR	$\mathbb{E}[Y(\bar{x})]$	outcome or treatment	augmented IPW

Two kinds of "model": *structural* (an assumption on the estimand: MSM form, SNMM blip; saturated means no assumption, italic above) and *nuisance* (a working model of the observed data, outcome or propensity). So MSM/SNMM carry an extra structural assumption that the g-formula and IPW do not. Mind the level: MSM and SNMM are models placed on an estimand; IPW, g-estimation and AIPW are estimators (MSMs are fitted by IPW, SNMMs by g-estimation). SNMM is the outlier, targeting a different estimand, the history-specific blip, which enables time-varying effect modification and optimal dynamic regimes.

The big picture. Treatment–confounder feedback ($X_0 \to Z_1 \to X_1 \to \cdots$) defeats standard methods. The g-formula standardizes over time-varying Z; IPW/MSM re-weights to remove $Z \to X$ arrows; g-estimation/SNM subtracts blip effects backward in time; doubly robust AIPW combines outcome and treatment models.

Key takeaways.

1. Time-varying treatments require extending the causal framework to multiple time points and treatment strategies.
2. When a time-varying confounder is affected by prior treatment, standard regression, stratification and matching all fail.
3. The g-methods handle this under sequential exchangeability, consistency and positivity.
4. Doubly robust methods add protection against misspecification.
5. In practice choose by problem structure, available models, and sensitivity to positivity violations.